

09g — Profiling Database Performance

Key Concepts

Term	Definition
Profiling	Identifying slow or inefficient queries to fix before they cause production issues
EXPLAIN / EXPLAIN ANALYZE	SQL command that shows how the DB executes a query and where time is spent
Seq Scan	Sequential (full table) scan — reads every row — bad on large tables
Index Scan	Uses an index to find rows directly — fast
Slow Query Log	DB logs queries exceeding a time threshold — used to find worst offenders
APM	Application Performance Monitoring — continuously tracks query performance in production

The Fix Workflow

1. Identify slow query via slow query log or APM alert
2. Run `EXPLAIN ANALYZE` on it
3. Look for Seq Scans, missing indexes, expensive joins
4. Add index / rewrite query / add eager loading
5. Run `EXPLAIN ANALYZE` again to confirm improvement
6. Deploy and monitor

What to Look For in EXPLAIN Output

Symptom	Problem
<code>Seq Scan</code> on large table	Missing index
High row count scanned	Query not selective enough
Many repeated similar queries	N+1 problem
High cost on join step	Inefficient join / missing index on join column

Profiling Tools

Tool	Used For
<code>EXPLAIN ANALYZE</code>	Query execution plan — built into PostgreSQL/MySQL
Slow Query Log	DB-level logging of queries over threshold

Tool	Used For
MiniProfiler	.NET — query count and duration per request
EF Core Query Logging	Logs all generated SQL to console/file
Application Insights	Microsoft/Azure APM — tracks slow queries in production
New Relic / Datadog	Production APM — alerts on slow queries across all services
pgAdmin	Visual query analysis for PostgreSQL

APM vs Manual Profiling

	Manual (EXPLAIN)	APM Tools
When	On-demand investigation	Continuous, always-on
How	You run it manually	Automatically monitors all queries
Best for	Diagnosing a specific slow query	Catching regressions in production

Industry Examples

- **Shopify** — Datadog APM catches N+1 and slow queries before they reach customers
- **GitHub** — runs EXPLAIN on every slow query alert; indexes are first line of fix
- **Stack Overflow** — MiniProfiler built by their team for per-request query tracking in .NET

Interview Questions

Question	One-Line Answer
How do you find slow queries in production?	Slow query log + APM tools like Datadog or Application Insights
What does EXPLAIN ANALYZE tell you?	Execution plan — indexes used, rows scanned, time per step
What is a Seq Scan and why is it bad?	Full table scan — reads every row — very slow on large tables, usually means missing index
What tools have you used for DB profiling?	EF Core query logging and MiniProfiler in .NET
What is APM?	Application Performance Monitoring — continuously tracks query performance and latency in production